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Cleaning Ancient Coins
Some Suggestions.

Tools Techniques (bronze) Techniques (silver) Cleaning Silver Finishing Touches Preservation Storage A Note About Patina

Relax and put your feet up while we explore the world of cleaning ancient bronze or silver coins. In each section of cleaning techniques, I have listed a number of possibilities. With each method I have also included a brief critique and whether or not I can recommend it. You may want to try them all to find the one that is right for your circumstance.

Email Kevin if you have any questions.

Notes and Cautions:

- Not every coin needs to be cleaned!!
- When you first start, practice on a coin that you can afford to lose. When you try a new technique practice on a coin you can afford to lose. Bottom line is, when starting out you will ruin some coins. It is unfortunate but all part of the learning curve.
- The object of cleaning coins is *not* to reduce them to bare metal lumps. Your concern should be with removing dirt and encrustations and leaving the base coin alone. A coin with its patina intact is always worth more than a bare metal coin. Please see our section on patina for more information.
- When using chemicals, always follow the manufacturers instructions for use and in mixing solutions. Many of the chemicals you will use are harmless, but for safety sake always read and follow instructions.
- There is no one 'right' method for cleaning coins. What works for one coin may be an absolute mess for another. The best solution seems to be a combination of methods and experience. Experiment to find the conbinations that work right for you.
- Pirates may have buried treasure, but if you are expecting to find rare or unusual coins in your uncleaned coins you will probably be disappointed. It is far better to look at it as a numismatic adventure. Cleaning ancient coins can be just plain fun and that is the way it should be approached.

- The most valuable tool in your cleaning tool kit is not one you can buy in the store. That tool is *patience*.

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Tools:

As you gain experience in cleaning coins, you will find yourself building up a small arsenal of useful tools. Here are some suggestions about the tools that I have found useful.

- **A strong directional light.** I use a goose neck reading lamp with a track lighting bulb that gives a highly directional light. It allows me to position the light anywhere and direct the beam where needed most. A must for the attribution process.

- **Brushes.** A variety of brushes is useful. Include toothbrushes and stiff bristle brushes (plastic or fiberglass) in different sizes. If feel you must use metallic brushes, make sure they are brass, not harder metals like steel. *Do not use brillo, scouring brushes, Dremel or similiar tools.*

- **Small detail tools.** Toothpicks, dental tools, straight pins and sharpened bamboo sticks are all possibilities. They are invaluable for working in small areas such as in between the letters of an inscription.

- **Cotton swabs and Q Tips.** Good for applying solutions to selected areas of a coin. Also sometimes useful for cleaning lightly dirty coins.

- **Glass or plastic jars, preferably with lids.** For mixing and storing solutions. Some of the solutions you will be working with are best kept confined. You only have to spill a batch of used olive oil once to realize the truth to this.

- **A handy source of hot and cold water.** Good to have for mixing solutions and giving your coins a water rinse. For best results use distilled and de-ionized water rather than tap water. Tap water frequently contains chemicals or disolved minerals that can cause unpredictable results.

- **Stereo Microscope.** Although expensive, I cannot say enough good things about the use of stereo microscopes when cleaning coins. A 10x power microscope will allow you to see an entire follis sized coin with high enough resolution that it makes cleaning a pleasure. In many cases all you will need in the way of tools is a needle or straight pin. With a stereo microscope and a needle you can place force

precisely where you need to on a coin rather than on the entire surface. A decent microscope will run you $200 or more. Try to find one used, if you can.

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Cleaning Techniques (bronze):

Soap and Water	A simple to clean coins is plain soap and water. I will usually take a paper cup, put a few coins in it and fill it with soapy water. I will swirl it around for a bit then empty out the dirty water. Make sure the soap residue is gone by giving the coins a fresh water rinse with several water changes. After that, dry the coins thoroughly and see if additional cleaning is necessary. *No muss, no fuss, no problems. Provided you make sure the coins are **thoroughly dried.***
Distilled Water Soak	By far the simplest way to clean dirty coins is let them soak in distilled water. A water soak can run from a few hours to several days. Just check them periodically, give them a little light brushing with your favorite brush and repeat as nescessary. After that, dry the coins thoroughly and see if additional cleaning is necessary. *Easy enough. And you may be surprised how much dirt you can remove with this simple method.*
Olive Oil	**A.** Put the coins in a glass and fill the glass with olive oil until the coins are covered. Leave them in the olive oil for 3-4 days. The olive oil will penetrate the dirt and soften it. **B.** After the 3-4 days, take the coins out and pat them dry with a paper towel. Mix a batch of TSP (Tri-Sodium Phosphate...a cleaner available in most home repair or paint shops) with one teaspoon in warm water. Put the coins in the TSP solution for 5-10 minutes. This will remove the olive oil and some of the dirt. **C.** Rinse the coins in clean water to remove the TSP residue. **D.** Take an old toothbrush or stiff bristle brush (plastic, fiberglass, etc.)and gently brush away at the coins.

	E. Examine the coins. At this point, some of them will be clean and identifiable. The ones that are not should go back into the olive oil for another 3-4 days soaking. **F.** Some coins will have dirt in recesses, between the letters of the inscription for example. See the Finishing Touch section for tips on this. **G.** Repeat this process until you are satisfied with the result. I have left some coins soaking for weeks before I was satisfied with the result. *I have used this method myself, extensively, and have found it to be safe with no discernable drawbacks. Time consuming, but safe. If you cannot find TSP, then any industrial strength degreaser should work.*
Heavy Penetrating Oils.	Some people have reported good results in using WD40 or other heavy penetrating oils in place of olive oil. The penetrating oil will soak through the coin and gradually loosen incrustations. Use the penetrating oil the same way you would use olive oil. A series of soak, clean and rinse cycles seems to work best. *The only drawback that has been reported is a tendency to darken or change the color of the patina.*
Baking Soda and Vinegar	A combination of vinegar and baking soda will clean some lightly dirty coins. I will ,sometimes, use this method when soap and water does not work. The chemical reaction between the vinegar and baking soda will lift much of the dirt off. I recommend immersing the coins no more than 30 seconds at a time. More than that and the acidic reaction may start working on the coin itself. Make sure you rinse the coins in fresh water,making several water changes, after finishing. The object is to be sure the reaction has stopped. *A drawback to the vinegar/baking soda combination is that the patina on some coins may be lightened. Particularly coins with a green patina.* *An additional potential drawback to the vinegar and baking soda method has been noted. If the coin has subsurface pitting, then this method seems to draw it out more than other methods.*
Sonic or Ultrasound Cleaners.	Generally reported to be a waste of time. I have tried it with a small jewelry type ultrasound cleaner and have had mixed results. On some coins, it seems to make the dirt a little looser and therefore easier to remove. It does not actually clean the coins. Some customers have reported good results with

	'industrial' type ultrasonic cleaners. These machines are generally expensive and out of reach for the average person. Unless you can find one used. *If you have one at home, fine give it a shot. Otherwise it will cost you $50 and up to purchase one. If you really have to try it, see if you can borrow one first.*
Rock Tumblers.	Used for polishing gems and minerals. I have been skeptical of this method, but have received reports of good results. It may be useful in the initial cleaning stages when you are trying to remove the first few layers of dirt. Just place the coin in the tumbler with water and a little dish washing detergent and let it run for 5 - 10 minutes. Make sure you rinse the coins off with fresh water after to remove the detergent residue. *Personally, I would not use a rock tumbler for more than one coin at a time. It would seem to me that the risk of the coins damaging each other would be too high. I would be glad to hear from anyone who has a contrary opinion.*
Electrolysis.	I am currently investigating the effectiveness of cleaning by electrolysis. The good news is that the components that you need are generally available around the house and that it only takes 5 minutes to get set up. The bad news is that the technique requires working with electricity (either through a step-down transformer or batteries) and could be a hazard to children. My initial impression is that, judging by the color of the water after a session, electrolysis does seem to have some effect. I will post a more comprehensive evaluation at a later date. *Because of the potential hazard to children, I am holding off posting a complete description at this time.*
Calgon Water Softener.	Calgon is a liquid that can be added to soften hard water and it can be found in most supermarkets. It can also be used to remove hard mineral encrustations on bronze coins. Just mix a solution of Calgon and used to remove built up dirt. Make sure you rinse the coins off with fresh water after to remove any residue. *It works. But be careful. A coin left to soak for too can end up being attacked by the Calgon rather than being helped by it. May also be used in spot applications by dabbing a little bit with a Q-Tip on selected areas of your coin.*

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Cleaning Techniques (silver):

Lemon Juice	A relatively fast technique for a little light cleaning. Simply take some lemon juice (I have used the concentrate type found in most grocery stores) and allow the coin to soak for a few minutes. For fast cleaning I will use the concentrate straight from the bottle. If I want to go slower I will dilute it with distilled water. After three or four minutes take the coin out, dry and examine it. If the coin is clean to your satisfaction, give it a running water bath to remove any trace remains of the juice. Or let the coin soak in a dilute solution of baking soda and water. Repeat as necessary, but if the coin does not 'come clean' within a few cycles then it is not likely too. Note that as the lemon juice becomes exhausted it will turn an interesting shade of green. *A simple method for light cleaning. Does not work on all coins.*
Ammonia	A more corrosive technique than lemon juice, it can be tried for those hard to clean coins. To use simply follow the same steps you used for the lemon jouce soak. But be prepared to check the coin frequently. *A simple method for heavier cleaning. Does not work on all coins.*

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Finishing Touches

Once you are satisfied that a coin is about as clean as it will get, you will probably need to touch up some small areas on your coins or preserve it to keep it from corroding.

Small Area Touch Up

Small areas (for example between the letters of an inscription) can be cleaned with a steady hand, bright light and a variety of tools. Which tools you will use will depend upon your personal preferences and what you feel comfortable using. I have had a number of different tools suggested to me including:

- *dental tools*

- *sharpened bamboo sticks*
- *toothpicks*
- *a glue gun*

A special note should be made about the last one. Take a glue gun (the type that uses glue sticks and can be found in any arts and crafts store) and put a blob of glue on a coin. Let the glue harden and then remove it. The glue should be relatively easy to remove. If it works right, the glue will lift the dirt from the hard to reach areas. With a few reservations I can tell you that it works. When you pry the hardened glue off it pulls the dirt out of even the tiniest spot. It may not get all of it off with the first attempt so multiple applications may be needed. Make sure the glue has hardened before trying to remove it. The reservations are:

- If you are working on a 'silvered' coin the glue may take the dirt and the silvering off at the same time. So don't use it on that silvered antoninianus.
- If the coin has subsurface pitting, then the glue may expose the pits.

So, if you are careful about which coin you use it on, then this technique will work.

I would highly recommend that you practice on cheap coins first before working on the sestertius of Caligula that you've been dying to touch up. ;>

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Coin Preservation and Protection

After you have gone through all of the trouble to clean your coins, now you want to preserve and protect them. To do so, some collectors coat their coins to seal out moisture and the environment. Some traditional techniques from the last century and still in use in Europe are to lightly coat the coins with bees wax or varnish. New products are available which include:

- Blue Seal Coin Cleaner and Preservative. This product lightly cleans coins, coats (imparting a soft sheen) and seals them. I apply it with a QTip and wipe any excess off with a soft rag. It works, but has the drawback of slightly darkening the patina. Also, porous coins will soak it up like a sponge and darken considerably. The good side is that non-porous coins that already have a dark patina come out looking super. It is not for every coin and requires some care in its use. It is available in some numismatic supply stores. If you cannot find it, check the Book Seller List for information on contacting the **Brooklyn Gallery of Coins and Stamps**. They sell numismatic supplies as well as reference works.

- Renaissance Micro-Crystalline Wax Cleaner / Polish. This is a product, imported from England, that is used by the British Museum

for its collections. According to the description it "revives and protects, freshens colours and imparts a soft sheen. Dries instantly.". It is available in an 8oz tin from Art Rubino at **Numismatic Arts of Santa Fe**. If you want to give it a try, the phone number of Numismatic Arts is 505 982-8792 or email Art for availability and details. I have tried it and it works as advertised. Plus that 8oz tin will most likely last forever.

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Storage

There is not much I can say about storing your coins that hasn't already been said. The one point that I would like to repeat is that if you use 2x2s, flips or any storage product that uses plastic, make sure they do not contain PVC (polyvinyl chloride). PVC will do horrible things to your coins over a period of time and may ultimately ruin them.

Non-PVC storage products are available through most supply houses. One of my favorite sources is the **Brooklyn Gallery** in Brooklyn NY. They carry a nice selection of numismatic supplies including the non-PVC flips. BTW...in case any one is curious the name of my prefered flip is ET Safflip manufactured by the ET Kointain Company.

Art Rubino at **Numismatic Arts of Santa Fe** stocks Acid Free Paper Coin Supplies. They are "two by two" white paper, acid free envelopes for coins, and 24 mm [slightly less than one inch] in white only. The envelopes are white paper since they cannot guarantee acid free paper with the presence of any coloration. These are the very same supplies used by the British Museum in London for preservation of their extensive collections.

Contact him at 505 982-8792 or by email for current pricing information.

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A Note About Patina

PATINA *(noun), a thin film of corrosion, usually green, that forms on copper and bronze as a result of oxidation.*

Doesn't sound like much, does it ?
The patina of an ancient coin has taken literally centuries to form and has helped to protect the metal of the coin from the elements and further corrosion. An ancients coins patina can appear as green, brown, black and many shades in between. It is part of a coins history and as such, should be left as intact as possible.

Consider that coins with a poor or spotty patina are generally worth less than coins with a nice even patina. Furthermore, the value of a coin that has had

the patina removed can be severely reduced.

So, it is in your own interest as a collector and ancient coin cleaner to take care of your coins patina. Your objective when cleaning coins should be just to remove any dirt and incrustation that obscures the coins design and not to return them to the way they looked when they were first minted. Remember, *don't clean them down to the bare metal.* Ancients coins are not supposed to be bright and shiny.

Ancient silver coins can also have a form of patina on them, but we call that Toning. Toning can range from a very light to a very dark grey. A properly toned silver coin can be very pleasing to look at, as opposed to the bright silver coins that are so common today. So please, if you have a toned silver coin let it be.

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